

Programme	B. Sc. Computer Science				
Course Title	Exploratory Data Analysis				
Type of Course	Elective				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Basic Statistical Knowledge 2. Python Programming including knowledge in Pandas library				
Course Summary	This course explores the different visualization tools and techniques and teaches the application of these techniques using Python packages.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the importance of data visualization for business intelligence and decision making.	U	C	Instructor-created exams / Quiz
CO2	Understand different types of charts and plots such as line, area, histograms, bar, pie, box, scatter, and bubble.	U	C	Instructor-created exams / Quiz
CO3	Learn about categories of visualization and application areas.	R	C	Instructor-created exams / Quiz
CO4	Familiarize with the data visualization tools and techniques.	Ap	P	Assignments/ Case Studies
CO5	Familiarise with the Python libraries, such as Matplotlib, Seaborn, Folium, Bokeh and learn how to tell a stimulating story.	Ap	P	Assignments/ Case Studies
CO6	Create advanced visualizations for geo spatial data.	Ap	P	Assignments/ Case Studies

* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)
 # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)
 Metacognitive Knowledge (M)

Detailed Syllabus:

Module	Unit	Content	Hrs 48+12	Marks 70
I	Introduction to Data Visualization		15	20
	1	Data:- Types of Data-Structured and Unstructured Data, Qualitative and Quantitative Data, Continuous and Discrete Data, Primary and Secondary Data, Data Attributes - Types of Data Attributes - Nominal, Ordinal, Interval, Ratio	3	
	2	Introduction to Data Visualization:- Data Visualization, The Importance of Data Visualization, Overview of popular data visualization libraries in Python - Matplotlib, Seaborn, Folium, Bokeh	1	
	3	Plots:- Comparison Plots: Line Chart, Bar Chart and Radar Chart	2	
	4	Relation Plots: Scatter Plot, Bubble Plot, Correlogram and Heatmap	2	
	5	Composition Plots: Pie Chart, Stacked Bar Chart, Stacked Area Chart, Venn Diagram	3	
	6	Distribution Plots: Histogram, Density Plot, Box Plot, Violin Plot	2	
	7	Geo Plots: Dot Map, Choropleth Map, Connection Map	2	
II	Data Visualization with Matplotlib		10	20
	8	Introduction, Overview of Plots in Matplotlib Pyplot Basics: Creating Figures, Closing Figures, Format Strings, Plotting, Plotting Using pandas DataFrames, Displaying Figures, Saving Figures	3	
	9	Basic Text and Legend Functions: Labels, Titles, Text, Annotations, Legends	1	

	10	Basic Plots: Bar Chart, Pie Chart, Stacked Bar Chart, Stacked Area Chart, Histogram, Box Plot, Scatter Plot, Bubble Plot	3	
	11	Layouts: Subplots, Tight Layout, Radar Charts, GridSpec	2	
	12	Images: Basic Image Operations, Writing Mathematical Expressions	1	
III	Simplifying Visualizations using Seaborn		12	15
	13	Introduction, Advantages of Seaborn, Plot a Relation Plot, Line Plot, Box Plot and, a Heat Map	2	
	14	Controlling Figure Aesthetics: Seaborn Figure Styles, Removing Axes Spines, Contexts	2	
	15	Color Palettes: Categorical Color Palettes, Sequential Color Palettes, Diverging Color Palettes	4	
	16	Interesting Plots in Seaborn: Bar Plots, Kernel Density Estimation, Plotting Bivariate Distributions, Visualizing Pairwise Relationships, Violin Plots	4	
IV	Plotting Geospatial Data		11	15
	17	Introduction to Geoplotlib, The Design Principles of Geoplotlib	1	
	18	Geospatial Visualizations - Choropleth Plot, GeoJSON File	2	
	19	Introduction to Folium	1	
	20	Visualizing Data: Building a Google map from geocoded data	2	
	21	Making Things Interactive with Bokeh : Introduction to Bokeh, Concepts of Bokeh, Interfaces in Bokeh, Output	3	
	22	Bokeh Server, Presentation, Integrating, Adding Widgets	2	
V	Open ended Module Practical Applications - Implement any 10 programs		12	

		Comparison Plots: Line Chart, Bar Chart, and Radar Chart 1. Write a Python script to create a line chart comparing the sales performance of two products over different months using Matplotlib. 2. Create a bar chart using Seaborn to visualize the average scores of students in different subjects. 3. Implement a radar chart using Matplotlib to compare the performance of multiple candidates in different skills.	2	
		Relation Plots: Scatter Plot, Bubble Plot, Correlogram, and Heatmap 4. Generate a scatter plot using Seaborn to analyze the relationship between the height and weight of individuals in a dataset. 5. Create a line graph with bokeh using Annotations and Legends. 6. Plot a correlogram heatmap using Seaborn to visualize the correlation matrix of variables in a dataset.	2	
		Composition Plots: Pie Chart, Stacked Bar Chart, Stacked Area Chart, Venn Diagram 7. Implement a pie chart using Matplotlib to represent the distribution of expenses in a budget. 8. Create a stacked bar chart using Seaborn to visualize the sales performance of different product categories over multiple quarters. 9. Generate a stacked area chart using Matplotlib to display the cumulative distribution of COVID-19 cases over time in different regions. 10. Use the matplotlib-venn library to create a Venn diagram illustrating the intersection of sets in a survey dataset.	2	

		Distribution Plots: Histogram, Density Plot, Box Plot, Violin Plot 11. Write a Python function to generate a histogram using Matplotlib for analyzing the distribution of exam scores in a class. 12. Create a density plot using Seaborn to visualize the distribution of income levels in a population. 13. Implement a box plot using Matplotlib to compare the distribution of salaries across different job roles. 14. Generate a violin plot using Seaborn to compare the distribution of ages between male and female participants in a study.	3	
		Geo Plots: Dot Map, Choropleth Map, Connection Map 15. Use Folium to create a dot map representing the locations of earthquake occurrences around the world. 16. Generate a choropleth map using Folium to visualize the population density of different countries. 17. Create a connection map using Matplotlib to illustrate flight routes between various cities.	3	

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6
CO 1	-	-	2	2	2	2
CO 2	-	-	2	2	2	2
CO 3	-	-	-	2	2	2
CO 4	-	-	2	2	2	2
CO 5	-	-	2	2	2	2
CO 6	-	-	2	2	2	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓		✓	✓
CO 3	✓			✓
CO 4	✓	✓		✓
CO 5	✓	✓	✓	✓
CO 6	✓	✓		✓

References

1. Data Visualization workshop, Tim Grobmann and Mario Dobler, Packt Publishing
2. Kristen Sosulski, "Data Visualization Made Simple", Taylor & Francis, 2019.
3. Pooja, Dr. Data Visualization with Python: Exploring Matplotlib, Seaborn, and Bokeh for Interactive Visualizations. BPB Online, 2023.
4. Wilke, Claus O. Fundamentals of data visualization: a primer on making informative and compelling figures. O'Reilly Media, 2019.
5. VanderPlas, Jake. Python data science handbook: Essential tools for working with data. "O'Reilly Media, Inc.", 2016.

Online Learning Resources

1. <https://www.coursera.org/courses?query=data%20visualization>
2. <https://www.simplilearn.com/free-data-visualization-course-online-skillup>